Simple Guide: Distributed Food Ordering

Kafka (KRaft) + Spark Batch · Big Data 2026 (UE24CS343AB2) · Team of 4: Member 1 = VM1, Member 2 = VM2, Member 3 = VM3, Member 4 = VM4

Section	What's in it
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1. The whole assignment in one minute

A food delivery app has a file of **62 orders** (`order.jsonl`). Some of them are junk: broken records, cancelled orders, bad timestamps, duplicates. We have to:

- **Send** the raw orders through Kafka (VM1 -> VM4).
- **Clean** them and add delivery numbers with Spark (VM4). 62 raw -> **44 good orders**.
- **Send** the cleaned orders back through Kafka (VM4 -> VM1).
- **Summarise** them with Spark (VM1) into two reports: one per restaurant, one per city.
- **Prove** that Kafka keeps working when one broker (VM3) is switched off.

Analogy. Kafka is a **post office** that stores letters (messages) in labelled mailboxes (*topics*). It keeps two copies of every letter, one in each branch (VM2 and VM3), so if one branch closes the letters are still there. Spark is the **calculator** each side uses on its own desk after collecting the mail.

The 4 machines

Member / VM	Role	What it does in plain words
Member 1 / VM1	Producer 1, Consumer 2, Spark Job 2	Posts the raw orders into Kafka. Later collects the cleaned orders and makes the two final CSV reports.
Member 2 / VM2	Kafka broker 1 + controller	A Kafka server, and also the **boss** of the cluster: it knows which servers are alive and who is in charge of each piece of data.
Member 3 / VM3	Kafka broker 2	The second Kafka server. Holds a backup copy of everything. This is the one we switch off in the failure test.
Member 4 / VM4	Consumer 1, Spark Job 1, Producer 2	Collects the raw orders, cleans them and adds metrics with Spark, then posts the cleaned orders back into Kafka.

The flow, step by step

```
STAGE 1                                      STAGE 2
VM1  producer1.py                            VM4  producer2.py
      |  topic: orders  (62 msgs)                   |  topic: order_metrics (44 msgs)
      v                                             v
VM4  consumer1.py -> orders_landed.jsonl     VM1  consumer2.py -> order_metrics_landed.jsonl
      v                                             v
VM4  spark_job1.py -> order_metrics.jsonl    VM1  spark_job2.py -> restaurant_summary.csv
      (clean + enrich, 62 -> 44)                                  -> city_summary.csv

Both topics live on the Kafka cluster = VM2 + VM3 (connected over ZeroTier)
Each topic: 2 partitions, replication factor 2
```

Rules from the spec (the things examiners check)

- Spark never reads Kafka directly. A consumer first saves the data to a local `.jsonl` file; Spark reads that file.
- No HDFS. Everything is on the normal local disk (`outputs/`).
- VM4 must not copy its file to VM1. It must go through Kafka (topic `order_metrics`).
- Both topics: **2 partitions, replication factor 2**, same cluster.
- Failure test: stop **VM3**, never VM2 (VM2 is the only boss).
- Exact duplicates counted once in both Spark jobs. All calculated numbers rounded to 2 decimals. Nothing hardcoded.

2. Demo day overview

The demo is one sequence of steps, 0 to 22, run in order (section 3). It covers exactly what the evaluator grades. The members take turns: each step says who runs it. When you finish your step, say **"done"**, and the member with the next step starts.

Order of turns

Steps	Member	What happens
0	Everyone	Set up the terminal
1	Member 2	Start broker 1 + controller
2	Member 3	Start broker 2
3-4	Member 2	Check both brokers, create fresh topics
5-6	Members 1 + 4	Clear old outputs, check brokers are reachable
7	Member 1	Producer 1
8-10	Member 4	Consumer 1, Spark Job 1, Producer 2
11-13	Member 1	Consumer 2, Spark Job 2, verify (graded outputs)
14	Member 2	Show topics and ISR before the failure
15	Member 3	Stop broker 2
16	Member 2	Show ISR and leaders after the failure
17	Member 1	Producer 1 with VM3 down
18	Member 4	Consumer 1 with VM3 down (data still flows)
19	Member 3	Restart broker 2
20	Member 2	Show VM3 rejoined the ISR
21	Member 1	Producer 1 after recovery
22	Member 4	Consumer 1 after recovery (demo finished)

That's the full graded demo. Optional steps O1-O5 (end of section 3) rerun the Spark jobs with VM3 down; use them only if the evaluator asks to see the whole pipeline working during the failure.

What the evaluator checks

TA checks	Steps	Who shows it	Evidence on screen
Environment ready	0	Everyone	`echo $KAFKA_HOME $BS` prints the Kafka path and VM2:9092
Old outputs cleared	5	Members 1 + 4	`outputs/` emptied before the live run
Pipeline output (accuracy)	7-13	Members 1 + 4	62 -> 44 -> 6 + 4 rows; verify says OK. **Only these outputs are graded.**
Topic config and ISR before	14	Member 2	PartitionCount 2, ReplicationFactor 2, Isr 1,2
ISR / leaders after VM3 stops	15-16	Members 3 + 2	Leader 1, Isr 1 everywhere; 4 under-replicated
Pipeline continues with VM3 down	17-18	Members 1 + 4	published 62 / landed 124 through VM2 only
VM3 rejoins the ISR	19-20	Members 3 + 2	Isr back to 1,2; under-replicated list empty
Pipeline continues after recovery	21-22	Members 1 + 4	published 62 / landed 186

Before you start (everyone)

- `git pull` so every VM has the same code. ZeroTier connected; you can `ping` VM2's ZeroTier IP.
- Use a **Linux / WSL terminal**, not PowerShell. The prompt must look like `user@machine:...$`, not `PS C:\...>`.
- Do step 0 in **every** terminal you'll use.
- Never stop VM2. Don't edit `order.jsonl`.
- Spark's `WARN` lines are normal. Only a `Traceback` means a failure.
- Keep each command's output on screen until the evaluator has seen it.

3. Demo flow: every command, in order

Demo day: go down the list in order. Run only the steps tagged with your member number; when a step has your tag, the step before it has just finished.

Practising alone (one laptop): run **every** step top to bottom, whatever its tag. Where a step shows a "Practising alone" box, use that one. (`bash practice_alone.sh` runs a longer version automatically, including the optional steps.)

Copying: copy one grey box at a time. Every line in a box is a real command. A line ending in `\` continues on the next line, so copy both lines together. The broker addresses are already filled in: VM2 = `10.31.88.79:9092`, VM3 = `10.31.88.79:9094`.

> **For now VM2's machine also runs VM3's broker** (second broker on port 9094), because Member 3 hasn't joined yet. That machine runs steps 1, 2, 15 and 19 with the "Practising alone" commands, after changing `localhost` to its ZeroTier IP in `advertised.listeners` of `kafka/local/broker1.properties` and `broker2.properties`.

Part A: Get ready

Step 0 · EVERYONE · Set up the terminal (every terminal you use)

Member 1 (VM1):

```
cd /mnt/c/Users/Aryan/Desktop/BD-Assignment
source .venv-wsl/bin/activate
export KAFKA_HOME=~/kafka
export JAVA_HOME=/usr/lib/jvm/java-21-openjdk-amd64
unset SPARK_HOME HADOOP_HOME HADOOP_CONF_DIR PYSPARK_PYTHON PYSPARK_DRIVER_PYTHON
export B1=10.31.88.79:9092
export B2=10.31.88.79:9094
export BS=$B1
export BOOTSTRAP=$B1,$B2
export KAFKA_BOOTSTRAP_SERVERS=$B1,$B2
```

Members 2 and 3 (VM2, VM3):

```
cd ~/BD-Assignment
export KAFKA_HOME=~/kafka
export B1=10.31.88.79:9092
export B2=10.31.88.79:9094
export BS=$B1
export BOOTSTRAP=$B1,$B2
export KAFKA_BOOTSTRAP_SERVERS=$B1,$B2
```

Member 4 (VM4):

```
cd ~/BD-Assignment
source .venv/bin/activate
export KAFKA_HOME=~/kafka
export B1=10.31.88.79:9092
export B2=10.31.88.79:9094
export BS=$B1
export BOOTSTRAP=$B1,$B2
export KAFKA_BOOTSTRAP_SERVERS=$B1,$B2
```

Practising alone (one laptop, local cluster):

```
cd /mnt/c/Users/Aryan/Desktop/BD-Assignment
source .venv-wsl/bin/activate
export KAFKA_HOME=~/kafka
export JAVA_HOME=/usr/lib/jvm/java-21-openjdk-amd64
unset SPARK_HOME HADOOP_HOME HADOOP_CONF_DIR PYSPARK_PYTHON PYSPARK_DRIVER_PYTHON
export B1=localhost:9092
export B2=localhost:9094
export BS=$B1
export BOOTSTRAP=$B1,$B2
export KAFKA_BOOTSTRAP_SERVERS=$B1,$B2
```

Expect: the prompt starts with `(.venv-wsl)` (Member 1) or `(.venv)` (Member 4). Member 4: if Spark later fails with `'JavaPackage' object is not callable`, run the `unset ...` line from Member 1's box.

Step 1 · MEMBER 2 (VM2) · Start broker 1 and the controller

Demo day:

```
$KAFKA_HOME/bin/kafka-server-start.sh -daemon ~/broker1.properties
```

Practising alone (one laptop, local cluster):

```
./kafka/local/cluster.sh format
./kafka/local/cluster.sh start 1
```

Expect: no output. When practising alone: `started broker 1 on port 9092`. `format` is needed after every WSL restart and is safe to repeat.

Step 2 · MEMBER 3 (VM3) · Start broker 2

Demo day:

```
$KAFKA_HOME/bin/kafka-server-start.sh -daemon ~/broker2.properties
```

Practising alone (one laptop, local cluster):

```
./kafka/local/cluster.sh start 2
```

Expect: no output. When practising alone: `started broker 2 on port 9094`.

Step 3 · MEMBER 2 (VM2) · Check both brokers joined

```
$KAFKA_HOME/bin/kafka-broker-api-versions.sh --bootstrap-server $B1 | grep "id:"
```

Expect: two lines, `id: 1` and `id: 2`.

Step 4 · MEMBER 2 (VM2) · Create fresh topics

```
./kafka/topics.sh delete
sleep 5
./kafka/topics.sh create
```

Expect: both topics with `PartitionCount: 2`, `ReplicationFactor: 2`, Isr 1,2. This also resets the message counts, so do it before every practice run.

Step 5 · MEMBERS 1 + 4 (VM1 / VM4) · Clear old outputs

```
rm -rf outputs/*.jsonl outputs/*.csv outputs/failover_test outputs/recovery_test
mkdir -p outputs/failover_test outputs/recovery_test
```

Expect: no output. This deletes the previous run's files on purpose; the pipeline makes new ones.

Step 6 · MEMBERS 1 + 4 (VM1 / VM4) · Check the brokers are reachable

Demo day:

```
nc -zv 10.31.88.79 9092
nc -zv 10.31.88.79 9094
```

Practising alone (one laptop, local cluster):

```
nc -zv localhost 9092
nc -zv localhost 9094
```

Expect: `succeeded` twice. `Connection refused` means a broker isn't running: redo steps 1-2.

Part B: Main pipeline (these outputs get graded)

Step 7 · MEMBER 1 (VM1) · Producer 1: send the raw orders

```
python producer/producer1.py --input order.jsonl
```

Expect: `published 62 records`.

Step 8 · MEMBER 4 (VM4) · Consumer 1: save the orders to a file

```
python consumer/consumer1.py --output outputs/orders_landed.jsonl
```

Expect: `landed 62 records`.

Step 9 · MEMBER 4 (VM4) · Spark Job 1: clean and enrich

```
python spark/spark_job1.py --input outputs/orders_landed.jsonl \
    --output outputs/order_metrics.jsonl
```

Expect: `valid records: 48`, `exact duplicates removed: 4`, `enriched records written: 44`.

Step 10 · MEMBER 4 (VM4) · Producer 2: send the cleaned orders

```
python producer/producer2.py --input outputs/order_metrics.jsonl
```

Expect: `published 44 records`, partition 0: 16, partition 1: 28.

Step 11 · MEMBER 1 (VM1) · Consumer 2: save the cleaned orders to a file

```
python consumer/consumer2.py --output outputs/order_metrics_landed.jsonl
```

Expect: `landed 44 records`, partition 0: 16, partition 1: 28.

Step 12 · MEMBER 1 (VM1) · Spark Job 2: build the reports

```
python spark/spark_job2.py --input outputs/order_metrics_landed.jsonl \
    --output-dir outputs
```

Expect: `6 rows -> restaurant_summary.csv` and `4 rows -> city_summary.csv`.

Step 13 · MEMBER 1 (VM1) · Verify and show the results

```
python tests/verify_outputs.py --raw order.jsonl \
    --metrics outputs/order_metrics_landed.jsonl
cat outputs/restaurant_summary.csv outputs/city_summary.csv
```

Expect: `OK: 44 enriched records ...` and the two tables from section 4.4.

Part C: Failure test, VM3 goes down

Step 14 · MEMBER 2 (VM2) · Show the topics before the failure

```
./kafka/topics.sh describe
$KAFKA_HOME/bin/kafka-topics.sh --bootstrap-server $B1 \
    --describe --under-replicated-partitions
```

Expect: Isr lists 1 and 2 on every partition; the second command prints nothing.

Step 15 · MEMBER 3 (VM3) · Stop broker 2 (the failure)

Demo day:

```
$KAFKA_HOME/bin/kafka-server-stop.sh
```

Practising alone (one laptop, local cluster):

```
./kafka/local/cluster.sh stop 2
```

Expect: the broker stops.

Step 16 · MEMBER 2 (VM2) · Show ISR and leaders after the failure

```
sleep 15
BOOTSTRAP=$B1 ./kafka/topics.sh describe
$KAFKA_HOME/bin/kafka-topics.sh --bootstrap-server $B1 \
    --describe --under-replicated-partitions
```

Expect: `Leader: 1` and `Isr: 1` on every partition; the second command lists all 4 partitions.

Steps 17-18 use `--bootstrap-servers $B1` **(VM2 only), to prove Kafka still works without VM3.**

Step 17 · MEMBER 1 (VM1) · Producer 1 with VM3 down

```
python producer/producer1.py --bootstrap-servers $B1 --input order.jsonl
```

Expect: `published 62 records`, so writes still work.

Step 18 · MEMBER 4 (VM4) · Consumer 1 with VM3 down

```
python consumer/consumer1.py --bootstrap-servers $B1 \
    --output outputs/failover_test/orders_landed.jsonl
```

Expect: `landed 124 records` (62 from the main run + 62 just sent), so reads still work. Say: "the data flow continues through the surviving broker".

Part D: Recovery, VM3 comes back

Step 19 · MEMBER 3 (VM3) · Restart broker 2

Demo day:

```
$KAFKA_HOME/bin/kafka-server-start.sh -daemon ~/broker2.properties
```

Practising alone (one laptop, local cluster):

```
./kafka/local/cluster.sh start 2
```

Expect: no output.

Step 20 · MEMBER 2 (VM2) · Show VM3 rejoined the ISR

```
sleep 15
./kafka/topics.sh describe
$KAFKA_HOME/bin/kafka-topics.sh --bootstrap-server $B1 \
    --describe --under-replicated-partitions
```

Expect: Isr lists 1 and 2 again; the second command prints nothing. Leader may stay 1 for a few minutes; that's normal.

Steps 21-22 have no `--bootstrap-servers` **flag: the commands use both brokers again.**

Step 21 · MEMBER 1 (VM1) · Producer 1 after recovery

```
python producer/producer1.py --input order.jsonl
```

Expect: `published 62 records`.

Step 22 · MEMBER 4 (VM4) · Consumer 1 after recovery

```
python consumer/consumer1.py --output outputs/recovery_test/orders_landed.jsonl
```

Expect: `landed 186 records`. **Demo finished.** When practising alone, stop the brokers with `./kafka/local/cluster.sh stop all`.

Why the numbers grow

	Main run (7-13)	VM3 down (17-18)	VM3 back (21-22)
`orders` topic / Consumer 1	62	124	186

Kafka keeps every message, so each producer run adds 62 more to the topic, and the consumer always reads the whole topic. Step 4 empties the topics again before the next practice run.

Optional, only if the evaluator asks: the whole pipeline with VM3 down

Run these **between step 18 and step 19** (VM3 still stopped). They show Spark and the second topic still work, and that the results don't change. They take about 2 minutes.

Step O1 · MEMBER 4 (VM4) · Spark Job 1 with VM3 down

```
python spark/spark_job1.py --input outputs/failover_test/orders_landed.jsonl \
    --output outputs/failover_test/order_metrics.jsonl
```

Expect: still `enriched records written: 44`, because the extra copies are exact duplicates.

Step O2 · MEMBER 4 (VM4) · Producer 2 with VM3 down

```
python producer/producer2.py --bootstrap-servers $B1 \
    --input outputs/failover_test/order_metrics.jsonl
```

Expect: `published 44 records`.

Step O3 · MEMBER 1 (VM1) · Consumer 2 with VM3 down

```
python consumer/consumer2.py --bootstrap-servers $B1 \
    --output outputs/failover_test/order_metrics_landed.jsonl
```

Expect: `landed 88 records` (44 old + 44 new).

Step O4 · MEMBER 1 (VM1) · Spark Job 2 with VM3 down

```
python spark/spark_job2.py --input outputs/failover_test/order_metrics_landed.jsonl \
    --output-dir outputs/failover_test
```

Expect: `exact duplicates removed: 44`, then 6 + 4 rows.

Step O5 · MEMBER 1 (VM1) · Show the results are unchanged

```
diff outputs/restaurant_summary.csv outputs/failover_test/restaurant_summary.csv
diff outputs/city_summary.csv outputs/failover_test/city_summary.csv \
    && echo "SAME RESULTS"
```

Expect: `SAME RESULTS`. Then continue with step 19.

4. Member 1 (VM1): Producer 1, Consumer 2, Spark Job 2

Member 1 runs three programs: **producer1.py** at the start, then later **consumer2.py** and **spark_job2.py** at the end. Two shared helper files sit behind them: `publish.py` and `landing.py`.

4.1 producer1.py: post the raw orders

Reads `order.jsonl` line by line and sends each line to topic **orders**. All of the real work is in the shared helper `publish.py`.

- **Sends every line as-is, junk included.** The producer only delivers; cleaning is Spark Job 1's job. If both did some cleaning, the rules would be split across two places.
- **No key.** To get a key (order_id) it would have to read the JSON, and some lines are broken. Without a key Kafka spreads messages randomly over the 2 partitions (e.g. 31 / 31, and the split changes each run).
- **acks="all"**: Kafka only says "received" once *every* in-sync copy has the message. This is the safest setting.
- **retries=10**: if something hiccups (e.g. VM3 goes down mid-send), it tries again automatically.
- **linger_ms=10**: waits up to 10 ms to bundle messages together, so fewer network trips.
- It waits for confirmation of every message (`flush()` + `f.get()`), so the printed "62 sent" means 62 *confirmed*, along with which partition each one landed in.

4.2 consumer2.py + landing.py: collect the cleaned orders

Reads *everything* from topic **order_metrics** and saves it to `order_metrics_landed.jsonl` (44 lines). Member 4's consumer1.py is the same code with topic **orders**.

- **Reads all partitions itself** (`assign()`, no consumer group). A consumer group remembers where it stopped, so a second run would get nothing. We want the *whole* topic every time.
- **Always starts from the beginning** (`seek_to_beginning`, offset 0).
- **Knows when to stop.** Normal consumers run forever. Ours stops once it has reached the last message of every partition *and* nothing new has arrived for 10 s (`--idle-timeout`).
- **Can start early.** `--wait-timeout` (default 120 s) lets it wait for the producer.
- **Overwrites the file** (`"w"` mode), so running it twice doesn't double the data.
- Prints how many records came from each partition: **16 from partition 0, 28 from partition 1**.

4.3 spark_job2.py: make the two reports

Reads `order_metrics_landed.jsonl` and writes **restaurant_summary.csv** (6 rows) and **city_summary.csv** (4 rows).

Step	What happens	Why
1. Read	Read the file with a fixed schema (types given up front)	Types are guaranteed and the file is read only once
2. Dedup	`dropDuplicates` on all 15 columns	Spec: exact duplicates count once. Removes 0 here because Job 1 already did it, but it protects against a producer being run twice
3. Group	groupBy restaurant_id, then groupBy city	The two analytics the spec asks for
4. Compute	count, averages, delayed count (rows where status = DELAYED), delay_rate_pct = 100 x delayed / count	delay_rate is worked out from the exact counts, and rounded only once at the end
5. Round	`F.round(..., 2)` on every calculated number	Spec rule

Step	What happens	Why
6. Sort + write	Sort the rows, collect them (just 10 rows) and write with Python's csv module	Spark's own writer makes a *folder* of part-files, but the spec wants one file with an exact name

`.cache()` keeps the deduplicated data in memory because it is used by two reports. Without it Spark would recompute it for each one.

4.4 Final outputs (know these by heart)

restaurant	orders	avg amount	avg prep	avg delivery	delayed	delay %
R01	8	548.5	13.13	22.0	4	50.0
R02	8	539.25	13.38	27.0	8	100.0
R03	7	664.0	13.86	32.0	4	57.14
R04	7	639.57	12.14	37.0	7	100.0
R05	7	615.14	12.0	42.0	3	42.86
R06	7	712.14	13.43	47.0	7	100.0

city	orders	avg amount	avg delivery	delayed	delay %
Bengaluru (R01+R02)	16	543.88	24.5	12	75.0
Hyderabad (R06)	7	712.14	47.0	7	100.0
Mumbai (R03+R04)	14	651.79	34.5	11	78.57
Pune (R05)	7	615.14	42.0	3	42.86

Easy cross-check: each restaurant is in exactly one city, so Pune = R05 and Hyderabad = R06. Bengaluru = R01 + R02: 8+8 = 16 orders, 4+8 = 12 delayed, 12/16 = 75%. Order counts add up to 44; delayed orders add up to 33 of 44. Example: R05 delay % = 100 x 3 / 7 = 42.857 -> **42.86**.

Your demo steps (section 3): 0, 5, 6, 7, 11, 12, 13, 17, 21 (optional: O3, O4, O5).
Show the evaluator: 62 published; 44 landed (16 / 28); 6 + 4 rows; verify OK; both CSVs; `published` 62 still works with VM3 down (step 17).
Be ready to explain: why Producer 1 has no key; why the consumer reads every partition from offset 0 and stops by itself; how Job 2 dedupes, groups and rounds; why producing still works with VM3 down (acks=all + min.insync.replicas=1).

5. Member 2 (VM2): Broker 1 + KRaft controller (the boss)

- Runs Kafka with `process.roles=broker,controller`, `node.id=1`.
- **KRaft** = Kafka manages itself, with no separate ZooKeeper. The controller keeps the cluster's "register" (which brokers are alive, which topics exist, who leads each partition).
- Port **9092** = normal Kafka traffic, port **9093** = controller traffic (only VM2 has it).
- Generates the **cluster ID** once; VM3 must use the same ID to join.
- `min.insync.replicas=1`: writes still succeed with only one copy alive (needed while VM3 is off).
- `auto.create.topics.enable=false`: topics are only created on purpose, so a typo can't create a wrong topic.
- In the demo, Member 2 shows the cluster state: both brokers, the topic config, and the ISR before, during and after the failure (section 8 explains how to read it).

Your demo steps (section 3): 0, 1, 3, 4, 14, 16, 20. You show the cluster state.
Show the evaluator: both broker ids; topics with 2 partitions and RF 2; Isr 1,2 before, then Leader 1 / Isr 1 after VM3 stops, then Isr 1,2 again after it restarts.
Be ready to explain: KRaft and what the controller does; node.id, listeners vs advertised.listeners, ports 9092/9093; cluster ID; min.insync.replicas=1; Leader vs Replicas vs Isr; why VM2 can't be stopped.

6. Member 3 (VM3): Broker 2 (the backup)

- Runs Kafka with `process.roles=broker` only, `node.id=2`.
- Points at VM2:9093 to join and is formatted with **VM2's cluster ID** (it must not create its own cluster).
- Holds the second copy of every partition. In the failure test, **this is the machine that gets stopped**.
- When it stops, VM2's controller notices the missing heartbeats, removes it from the ISR and makes broker 1 the leader everywhere. When it restarts, it copies what it missed and rejoins the ISR.

Your demo steps (section 3): 0, 2, 15, 19. Your broker is the one that fails.
Show the evaluator: the broker stopping and starting again. Then point to Member 2's describe output, where Isr shrinks to 1 and grows back to 1,2.
Be ready to explain: what happens step by step when VM3 stops (heartbeats stop, it's fenced, leadership moves to broker 1); how it catches up and rejoins the ISR; why VM3 uses VM2's cluster ID; what RF 2 protects against.

7. Member 4 (VM4): Consumer 1, Spark Job 1, Producer 2 (the cleaning step)

- **consumer1.py**: same code as consumer2, but reads topic **orders** -> `orders_landed.jsonl` (62 lines).
- **spark_job1.py**: the heart of the project. It **throws away bad records**, **removes duplicates**, and **adds 7 new columns**. Output: `order_metrics.jsonl` (44 lines).
- **producer2.py**: posts those 44 lines to topic **order_metrics**, **keyed by order_id**. The same key always goes to the same partition, so the split is always **16 / 28**.

Cleaning funnel: 62 -> 48 -> 44

Thrown out because...	Count
status is not DELIVERED (T04X0001-6: cancelled, pending, failed, refunded)	6
no order_id (2 "invalid_order_record" lines)	2
missing fields ({"order_id":"BROKEN"} and a T040002 copy without a city)	2
bad timestamp (T040005 "bad", T040003 "not-a-timestamp")	2
distance_km <= 0 (T040001: -2) and order_amount <= 0 (T040004: 0)	2
Total dropped -> 48 valid	**14**
Exact duplicates removed (T040006, T040017, T040028, T040039) -> **44**	4

New columns Spark Job 1 adds (all in minutes, rounded to 2 decimals):

Column	Formula
preparation_time_min	picked_up - accepted
delivery_time_min	delivered - picked_up
fulfillment_time_min	delivered - order_timestamp
delivery_speed_kmph	distance_km / (delivery_time_min / 60)
delay_min	max(0, delivered - expected)
delivery_status	ON_TIME if delay = 0, else DELAYED
delay_category	0 ON_TIME | up to 15 SLIGHT | up to 30 MODERATE | over 30 SEVERE

Midnight trick: it uses full date+time, not just the time of day. T040023 picked up 23:20 and delivered 00:02 the next day = **42 min**, not -1398. The time zone is set to UTC so daylight saving can't shift anything.

Your demo steps (section 3): 0, 5, 6, 8, 9, 10, 18, 22 (optional: O1, O2).
Show the evaluator: 62 landed; the drop report (14 dropped, 48 valid, 4 duplicates, 44 written); 44 published, split 16 / 28; `landed 124` with VM3 down and `landed 186` after recovery.
Be ready to explain: each cleaning rule and the 62 -> 48 -> 44 funnel; the 7 new columns and the midnight case (T040023 = 42 min); why Producer 2 is keyed by order_id; why the consumer lands 124 and 186 (Kafka keeps old messages).

8. The failure test explained (whole team)

```
Before (both up):    orders P0  Leader: 2  Replicas: 2,1  Isr: 2,1
                     orders P1  Leader: 1  Replicas: 1,2  Isr: 1,2
After stopping VM3:  orders P0  Leader: 1  Replicas: 2,1  Isr: 1     <- leader moved to 1
                     orders P1  Leader: 1  Replicas: 1,2  Isr: 1
Restart VM3 -> Isr back to 2,1 / 1,2.
```

Leader = the broker currently serving that partition. **Replicas** = where copies are *supposed* to live (doesn't change). **Isr** = copies that are currently up to date (shrinks to just 1 while VM3 is off).

Producing still works with VM3 down because the producer uses `acks=all` (wait for every *in-sync* copy, now just broker 1) and the brokers allow `min.insync.replicas=1`.

9. Key words in one line each

Term	Simple meaning
Broker	A Kafka server that stores messages (VM2, VM3).
Controller	The boss broker that tracks who is alive and picks leaders (only VM2).
KRaft	Kafka's built-in way of electing and storing the boss's records. It replaced ZooKeeper.
Topic	A named mailbox: *orders*, *order_metrics*.
Partition	A topic is split into pieces (we use 2) so work can be shared. Order is kept only inside one partition.
Offset	A message's position number inside a partition (0, 1, 2 ...).
Replication factor 2	Every partition is stored twice, once on each broker.
Leader / follower	The leader copy handles reads and writes; the follower copies it.
ISR	In-Sync Replicas: the copies that are fully up to date.
acks=all	The producer gets "OK" only when all in-sync copies have the message.
Bootstrap servers	The first broker address(es) a client contacts to discover the whole cluster.
advertised.listeners	The address a broker tells clients to use. It must be the ZeroTier IP, not localhost.
ZeroTier	A virtual LAN, so the 4 laptops on different home networks can reach each other.
local[*]	Spark runs on this one machine, using all CPU cores.
Lazy evaluation	Spark only plans the work; it runs when an action (count, collect) is called.

10. Top viva questions, short answers

Q. Why use Kafka instead of just copying files?

It separates the stages (VM1 and VM4 never talk directly), stores the data reliably in two copies, and lets any stage re-read the data. The spec also forbids copying the file.

Q. Why is Producer 1 unkeyed but Producer 2 keyed?

Raw lines may be broken JSON, so Producer 1 doesn't read them to find a key. Cleaned lines are always valid, so Producer 2 keys by order_id, which gives a fixed partition split (16/28).

Q. Why doesn't the consumer use a consumer group?

Groups remember where they stopped, so a second run would read nothing. Ours is a batch job that must copy the whole topic every time, so it reads all partitions from offset 0 and commits nothing.

Q. How does the consumer know when to stop?

When its position equals the last offset of every partition and nothing new has arrived for 10 s.

Q. What happens when VM3 stops?

VM3 stops sending heartbeats, the controller on VM2 removes it from the ISR and makes broker 1 the leader everywhere. Broker 1 already has a full copy (RF 2), so no data is lost.

Q. Why stop VM3 and not VM2?

VM2 is the only controller. Without it nobody can elect leaders. Real high availability needs 3 controllers.

Q. Can you still produce while VM3 is down?

Yes. acks=all only waits for the in-sync copies (now just broker 1), and min.insync.replicas=1 allows that.

Q. What if Producer 1 runs twice?

The topic has 124 messages, but Spark Job 1 still outputs 44 because duplicates are exact and get removed. Delivery is at-least-once, which is why both jobs dedupe.

Q. Why collect() and write the CSV with Python?

Spark's writer makes a folder of part-files. The spec wants single files with exact names, and the results are tiny (6 and 4 rows).

Q. Why is dropDuplicates in Job 2 if it removes 0?

The spec says both jobs must dedupe. It also protects against Kafka redelivery or a producer re-run.

Q. Why not deduplicate by order_id?

The spec says *exact* duplicates. Choosing between two different records with the same ID would be inventing a business rule.

Q. How do you know the outputs are correct?

tests/verify_outputs.py redoes everything in plain Python (no Spark) and compares every value. It prints "OK: 44 enriched records and both summaries match".

Q. Why Spark for only 62 records?

The assignment is about the architecture. The same code would work on millions of rows or on a real cluster.

11. Numbers to remember

Item	Value
Raw lines	62
Dropped by cleaning	14 -> 48 valid
Exact duplicates removed	4 -> 44 enriched
orders partition split	random, e.g. 31 / 31
order_metrics partition split	always 16 / 28
Restaurants / cities	6 (R01-R06) / 4 (Bengaluru, Hyderabad, Mumbai, Pune)
Delayed orders	33 of 44 (11 on time)
Delay categories	ON_TIME 11, SLIGHT 11, MODERATE 15, SEVERE 7
Highest / lowest delay %	100% (R02, R04, R06, Hyderabad) / 42.86% (R05, Pune)
Fastest / slowest delivery	R01 22.0 min / R06 47.0 min
Node IDs / ports	VM2 = node 1 (9092 + 9093); VM3 = node 2 (9092)
Versions	Kafka 4.3.1 (KRaft), PySpark 4.2.0, kafka-python 2.2.15, Java 17+